Quantum Walk-Based American Option Pricing vs Dynamic Programming

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Motivation

American-style options allow early exercise at any time before expiry. Optimal pricing requires solving a stochastic dynamic program over the full price trajectory.

Stock price model: $X_{k+1} = X_k + W_k$
where $W_k \sim e^2 - 0.5$

Profit at exercise: $(x - K)$ *(exercise only if $x \geq K$)*

Classical DP yields exact thresholds but scales poorly. Quantum walks offer a hardware-native path to stochastic transition estimation that may scale better.

Classical Dynamic Programming

Bellman Recursion:

$$V_m(x) = \max(x - K, E[V_{m-1}(x + W_{m-1})])$$

Key structural properties:

$V_m(x) - x$ is non-increasing in x
$V_m(x)$ is decreasing in m (time remaining)

Optimal threshold policy:

Exercise on day m iff $x \geq p_m$

$p_0 \geq p_1 \geq \cdots \geq p_N$
(thresholds decrease over time)

Computing exact thresholds requires the full transition kernel over the price grid, which becomes expensive as the number of bins grows.

Quantum Walk Approach

We replace the classical transition kernel with a quantum walk-sampled estimate, then feed it into the same Bellman recursion to derive exercise thresholds.

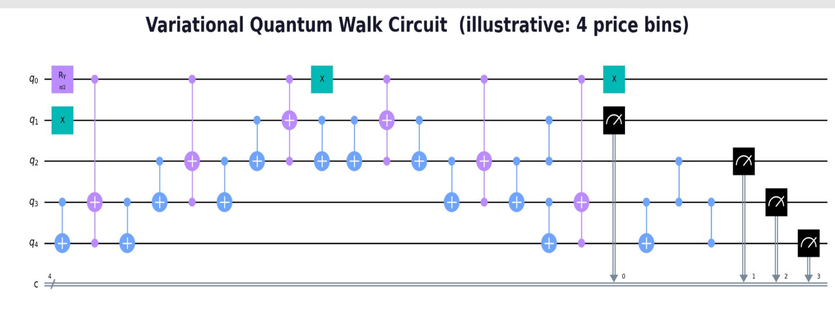
***Encoding:* Unary encoding, one qubit per price bin**

***Coin qubit:* $|0\rangle$ = move left (price down), $|1\rangle$ = move right (price up)**

***Walk step:* $U = S(C \otimes I)$,**
***Coin:* $C = R_y(\pi/2)$**

Quantum interference concentrates probability mass at the boundary bins, yielding a transition matrix T that differs structurally from the classical exponential random walk.

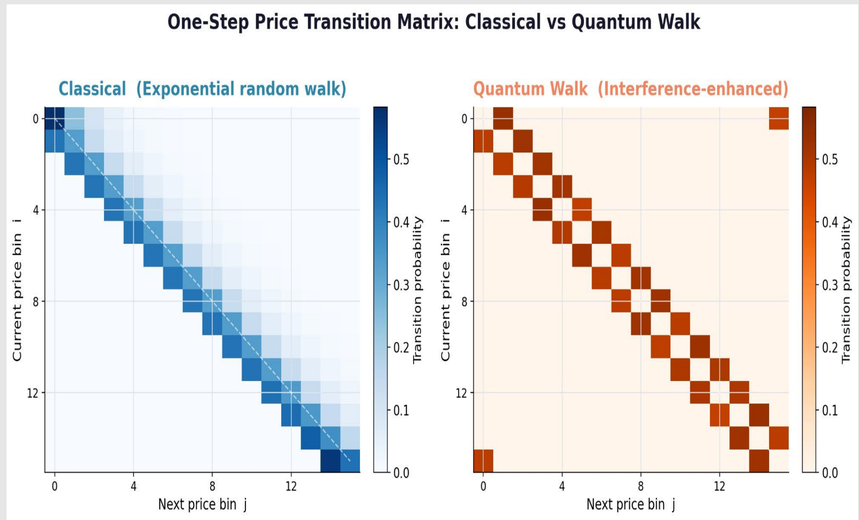
Quantum Walk Circuit (4 price bins)



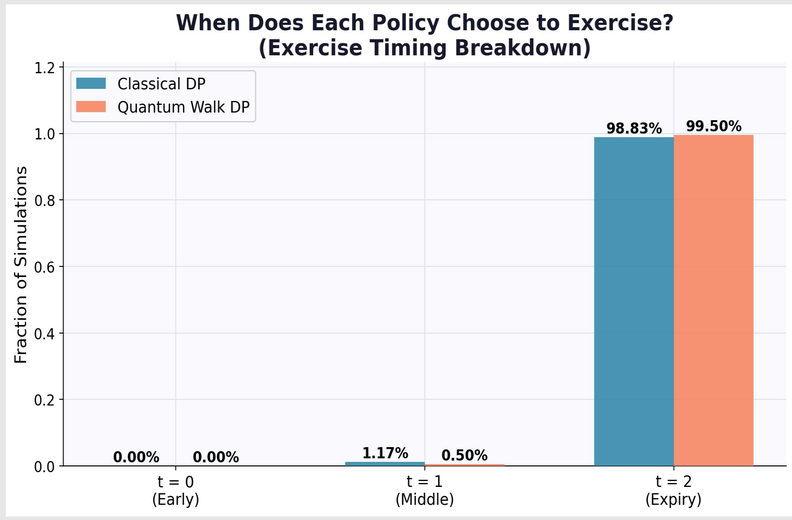
Future Work

- **Scale to larger price grids:** Test whether quantum advantage materializes as state space grows beyond classical tractability.
- **Real hardware execution:** Run on IBM or IonQ hardware to quantify noise effects on threshold estimation.
- **Multi-asset extension:** Explore entangled multi-coin walks for correlated asset pricing problems.

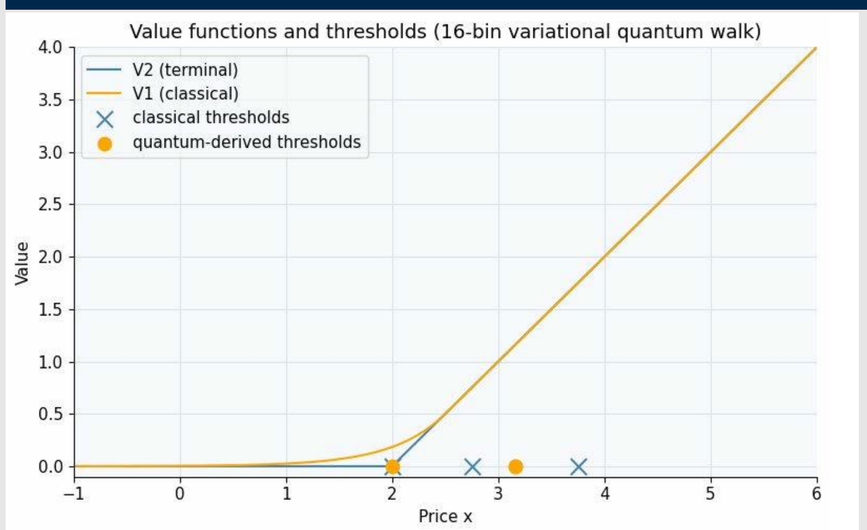
Price Transition Matrix: Classical vs Quantum

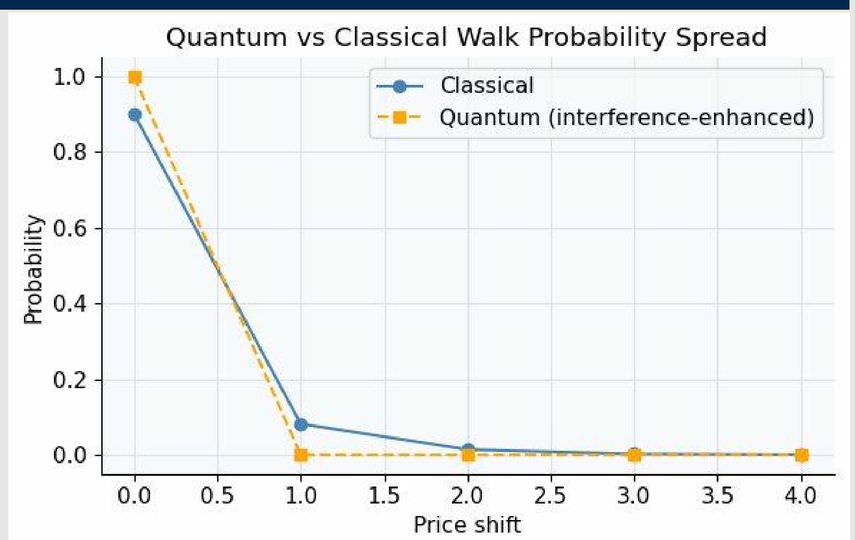


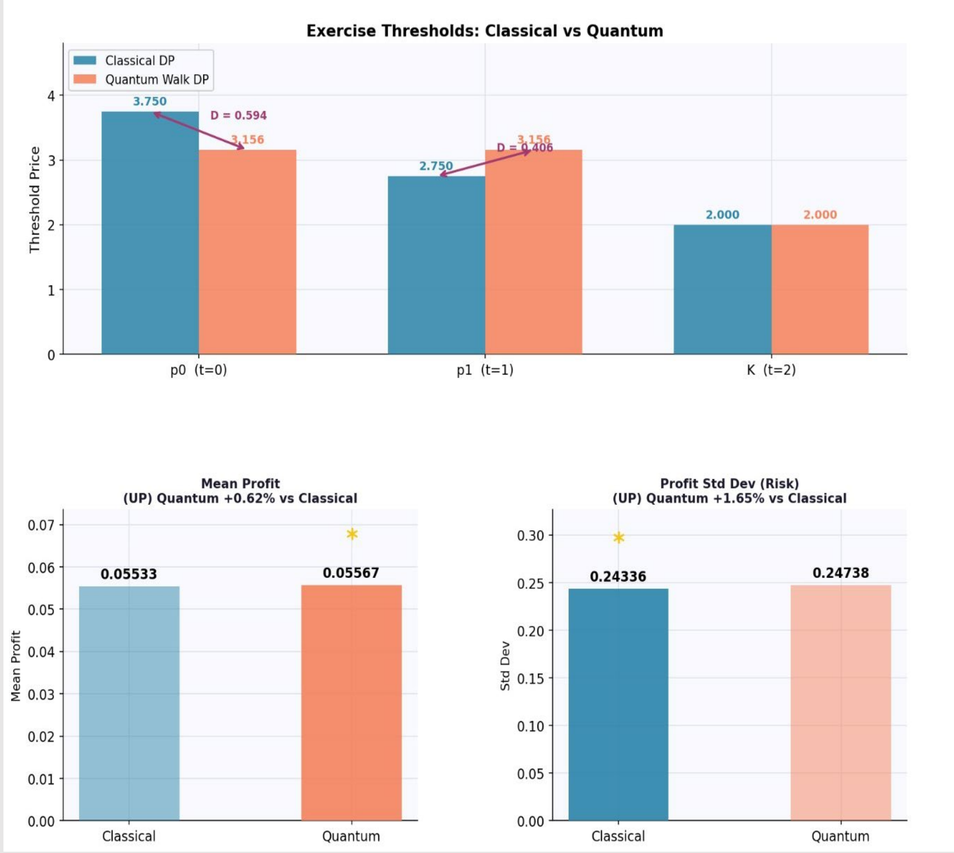
Exercise Timing by Policy



Results: Error & Performance Dashboard







↑ +0.62%
Quantum mean profit

-0.07%
Exercise fraction

Feature	Classical DP	Quantum Walk DP
Threshold calculation	Deterministic, exact	Estimated via quantum sampling
Mean profit	0.05533	**0.05567**
Std profit	0.24336	**0.24738**
Prop exercised	0.09058	**0.09052**
Early exercise behavior	Conservative	Slightly aggressive
Computational cost	Exact, potentially slower	Depends on quantum simulation

Conclusions

- **Policy agreement:** Quantum walk-derived thresholds closely match classical DP. Both policies exercise at expiry (t=2) in over 98% of simulations.
- **Performance parity:** Mean profit 0.05533 (Classical) vs 0.05567 (Quantum), Exercise proportion differs by less than 0.07%.
- **Threshold error:** 15.8% at t=0, 14.8% at t=1, 0% at expiry. The quantum walk recovers the qualitative shape of the optimal policy despite its differing transition structure.
- **Key insight:** The Bellman recursion structure is robust to moderate perturbations in the transition matrix, making quantum-sampled kernels viable for option pricing.

References

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